PROGRESS REPORT

Progress Results of the *Podostemum ceratophyllum* Transcriptome Analysis

Screening of associated-organism sequences, orthology with *Arabidopsis*, and root–leaf expression differences

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ROOT AND LEAF LIBRARIES

Root root part of the plant

PR1 | PR2 | PR3

> Split from the same plants · pairing of library numbers to be confirmed

Leaf all parts remaining after root removal

PL1 | PL2 | PL3

Current results at a glance

> The main computations are complete; identification ranks of bacteria, fungi and algae and the “unknown” sequences have been re-verified in targeted analyses.

	Analysis	Result
1	**Remove associated-organism reads before assembly and build the formal reference transcriptome**	Retention check of key markers and read-set sensitivity check completed
2	**Identify the sources of non-plant reads**	Mainly animals, plus bacteria, fungi and several algal groups; high-fragment sequences re-verified one by one for the reportable rank
3	**Proteome-wide correspondence with *Arabidopsis***	Proteome-wide correspondence table; root–leaf expression differences
4	**Targeted analyses of “unknown” sequences and the secondary cell wall**	Estimated host share of “unknown” sequences; expression of preselected secondary-wall genes

S1–S15 at the top right refer to sections of the Supplementary Data; “Source” at the bottom gives section and table numbers in the Comprehensive Experimental Record Report (the Report); [numbers] are references, listed on slides 36–38.

Source Report Table 1; 5.1, Table 99

Supplement S1

Sampling: root and leaf split from the same plants

Method Poly(A) mRNA libraries, Illumina NovaSeq X Plus PE150; raw-read QC with fastp 1.1.0, FastQC 0.12.1 and MultiQC.

> Root and leaf were split from the same plants and sequenced in one run; all comparisons in these slides use these six libraries.

Fraction	Sampled material	Libraries	Limits in comparisons
Root PR1–3	Root part of the plant	3	Split from the same plants as leaf; pairing and independent sampling units to be confirmed
Leaf PL1–3	All parts of the same plant remaining after root removal	3	Includes all non-root structures; not pure leaf blades

- The root–leaf test currently uses six libraries with the model ~ tissue, without same-plant pairing.
- A paired test will follow once the PR–PL correspondence and the independent sampling units are confirmed; heatmaps group libraries by tissue and do not treat PR1–PL1 etc. as confirmed pairs.
- Three unsplit whole-plant libraries (another sequencing run) went into the assembly but are not compared, because group and sequencing batch would be completely confounded [32]; the few pooled values that still include them are marked.

SEQUENCING RUN AND READ FEATURES

Root and leaf were sequenced in the same run (LH01101 run 29).

	Root	Leaf
Mean GC	47.50–48.13%	44.35–45.41%
Duplication	74.80–79.25%	70.71–78.43%

The lower GC of leaf mainly reflects a larger load of low-GC associated RNA, in leaf mostly animal rRNA such as that of Simuliidae (Report 4.2.2).

Source Report 4.1.1–4.1.2, Tables 2–3; 4.2.1, Table 5

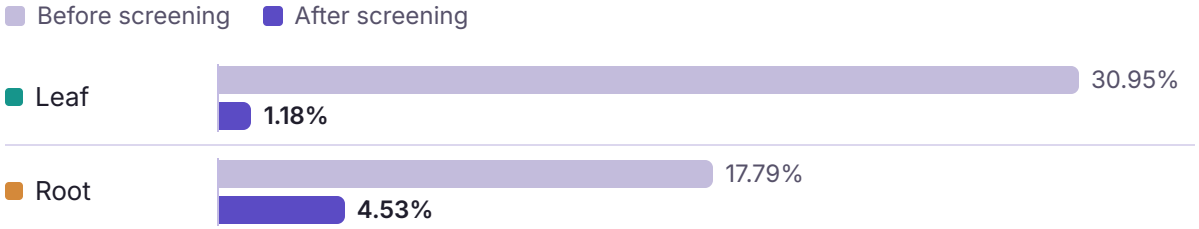
Supplement S1

Source screening: associated reads removed, plant signal stable

Method Rcorrector 1.0.7 and Trimmomatic 0.39; reads labelled on a first-round assembly (DIAMOND vs NR, BLASTN vs nt) and by Kraken 2, host reads rescued; Trinity 2.15.2; BUSCO 5.8.3.

> Most associated-organism reads were removed before assembly; the plant fragment share and the key expression results were not affected.

Non-plant fragments in the original library (same method; three libraries pooled)



- Reads removed: about 18% of all clean reads in root and 31% in leaf.
- With the labels of the three assembly versions unified, the formal assembly gives root 3.94% and leaf 1.10%.
- The plant fragment share of the original library barely changes with screening (root 65.0 / 65.1%, leaf 58.5 / 60.4%); what was removed is essentially the non-plant part.

99.6% of strict differential expression calls agree when all clean reads are used for quantification

27 marker genes with unchanged read counts

89.9% BUSCO completeness of the formal assembly unigenes (91.4% over all isoforms) [3]

- Of the 56 genes whose call changed, 46 gained reads from screened-out associated reads (fungus-, diatom-, green-alga- or animal-like reads mis-mapped to plant homologs); 8 had no removed reads, the change coming from normalization; 2 involved only ambiguous reads.

Source Report 4.2.4–4.2.5, Tables 12, 14; 4.2.7, Table 17; 4.2.10, Tables 22–24; 4.4.4, Table 43; 4.4.10, Tables 63–64

Supplement S2

What the non-plant RNA in the transcriptome is made of

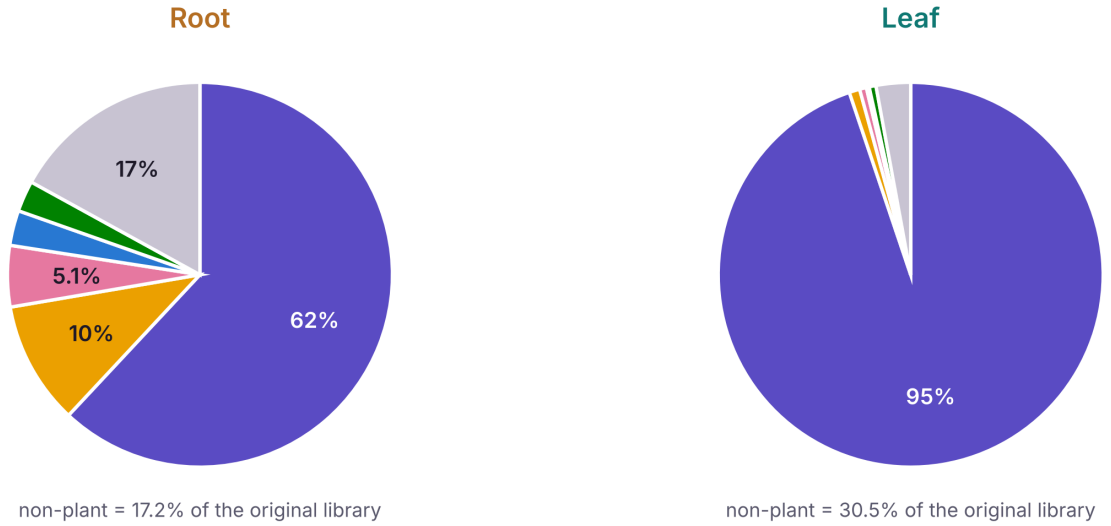
Method Salmon fragments on the first-round assembly (all clean reads); gene sources from NR / nt best hits and their common ancestor; 386 genes carrying most fragments of each group re-verified one by one; host isoforms deducted.

> Non-plant fragments are 17.2% of the root and 30.5% of the leaf original library. Animals make up 62% of them in root and 95% in leaf.

- **Root**: besides animals (mainly Chironomidae), diatoms and other stramenopiles are 10.3% of the non-plant fragments, fungi 5.1%, bacteria 2.9% and red and green algae 2.6%; the remaining 17.0% are mainly other eukaryotes, viruses and unresolved lineages (footnote a in the figure).
- **Leaf**: almost all non-plant RNA is animal, mainly Simuliidae (black flies, 20.8% of the leaf library).
- Bacteria, fungi and algae (including diatoms) together are 21% of the non-plant fragments in root (3.6% of the library) and 2.3% in leaf (0.7% of the library).
- These are shares of Poly(A) library RNA, not biomass or cell numbers; the next slide gives all thirteen major groups.

What the non-plant RNA in the transcriptome is made of

Share of all non-plant fragments in the original library (first-round assembly, all clean reads; mean of three replicates; re-verified labels, host organelle and rDNA isoforms removed). Poly(A) library RNA, not biomass or cell numbers.



non-plant = 17.2% of the original library

non-plant = 30.5% of the original library

	Root	Leaf		Root	Leaf
Animals	62.0%	94.8%	Bacteria	2.9%	0.2%
Diatoms and other stramenopiles	10.3%	0.9%	Red and green algae	2.6%	0.6%
Fungi	5.1%	0.6%	Other [a]	17.0%	2.9%

[a] Other eukaryotes, viruses, lineage unresolved, alveolates, Rhizaria, Amoebozoa and archaea.

Source Report 4.5.8, Table 95; Supplementary Data S2

Supplement S2

Sources of non-plant RNA fragments

Method Salmon fragments per gene on the first-round assembly (all clean reads); labels re-verified with nt and rRNA databases; host isoforms deducted.

> Non-plant fragments are mainly animal; the five bacterial, fungal and algal major groups make up about 3.63% of the root original library, and most are higher in root.

% of original library (mean of three replicates; re-verified labels)

Major group	Root	Leaf
Animals (Metazoa)	10.67	28.94
Stramenopiles (incl. diatoms)	1.782	0.271
Fungi	0.887	0.175
Bacteria (incl. cyanobacteria)	0.507	0.073
Red algae [b]	0.398	0.112
Green algae etc.	0.051	0.062
Other [a]	2.93	0.88
Non-plant total	**17.23**	**30.51**

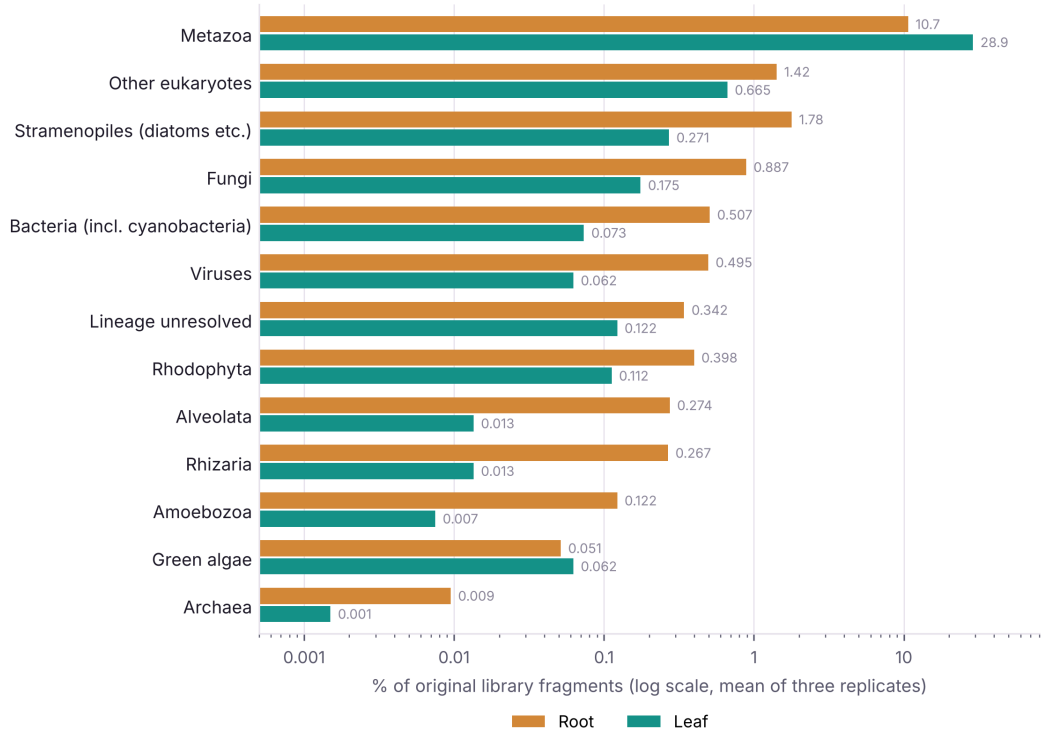
[a] Other eukaryotes 1.42 / 0.66, viruses 0.49 / 0.06, lineage unresolved, alveolates, Rhizaria, Amoebozoa, archaea. [b] Includes DN7210 (slide 14). The five bacterial, fungal and algal groups together: root 3.626%, leaf 0.693% (archaea and other eukaryotes excluded).

DEFINITIONS

- Shares of RNA fragments in Poly(A) libraries, not cell numbers, biomass or community abundance; Poly(A) selection misses most bacterial RNA. Stramenopiles are not all diatoms.

Non-plant RNA fragments by lineage

First-round assembly, all clean reads; host organelle and rDNA isoforms removed
Fragment shares of Poly(A) libraries, not biomass or cell counts



Source Report 4.5.1, Table 68 (original labels); 4.5.8, Table 95; re-verified values of all major groups in Supplementary Data S2

Supplement S2

Non-plant RNA fragments: changes across three label versions

Method V0 protein labels (DIAMOND vs NR) → V1 nt revision (megablast; common ancestor, identity-capped ranks) → sequence-by-sequence re-verification and isoform deduction.

> Changes in the totals across the three label versions are concentrated in a few high-fragment sequences (the original labels of script 84 sum to root 17.81% and leaf 30.91%; mean of three replicates).

- An rRNA-like sequence first labelled fungal is a bdelloid rotifer (Philodinidae, animal), about 0.28 / 0.03% (root / leaf); script 88 moved several more fungal labels to other eukaryotes or stramenopiles (one to Bacillariophyceae). Fungi in root fell from 1.225% to 0.887%.
- A sequence first labelled fungal is a grass (Poaceae, a non-host plant), about 0.036% in root (almost all PR3) and near 0 in leaf, and was removed from non-plant; script 88 also removed three green-alga-labelled embryophyte sequences (together root 0.001%, leaf 0.003%).
- A sequence first labelled stramenopile and moved to red algae by script 86 (closest reference *Palmaria*, 93%) has 97–99.9% of its fragments on host plastid rRNA isoforms and is recorded as host; leaf red algae fell by about 0.11 percentage points.
- DN7210, an 18S sequence first labelled alveolate and placed only in Eukaryota by script 86 (closest reference *Audouinella*, 90.4%), was reassigned on 2026-10-28 to red algae (Florideophyceae) from the common ancestor of named references; root 0.076%, leaf 0.067% (slide 14).
- A sequence first labelled other eukaryote is a mite (animal), about 0.05% in leaf.
- Viruses, Rhizaria and Amoebozoa are unchanged; the rise in animals comes mainly from the rotifer and mite reassignments.

HOST FRAGMENTS DEDUCTED BY ISOFORM

All 275,497 isoforms of non-plant genes were compared with host organelle and nuclear rDNA references; only 5 met the host standard, one in each of 5 gene clusters. Four carry many fragments: two “bacteria” (DN61527, DN2273; slide 9), the Chironomidae 28S cluster DN2745 (slide 8) and the “red alga” cluster DN22134; the fifth is a trace sequence with a bacterial label.

% of original library; the table on slide 6 shows values after deduction

	Root	Leaf
Host-isoform fragments of the 5 clusters	0.55	0.40
Before deduction: non-plant total	17.77	30.91
Before deduction: bacteria	0.635	0.331
Before deduction: red algae (incl. DN7210)	0.483	0.220

Source Report 4.5.2, Tables 70–72; 4.5.8, Tables 94–95

Supplement S2

Animal sources at a glance

Method Mitochondrial and rRNA marker genes vs nt (megablast); common ancestor of hits within 1%; ranks capped by identity (genus 97%, family 90%, order 85%).

> Animals account for most non-plant fragments; this round focuses on bacteria, fungi and algae, so for animals only family-level composition is reported.

% of original library, mean of three replicates; after nucleotide-based taxonomic revision and isoform-based deduction of host fragments

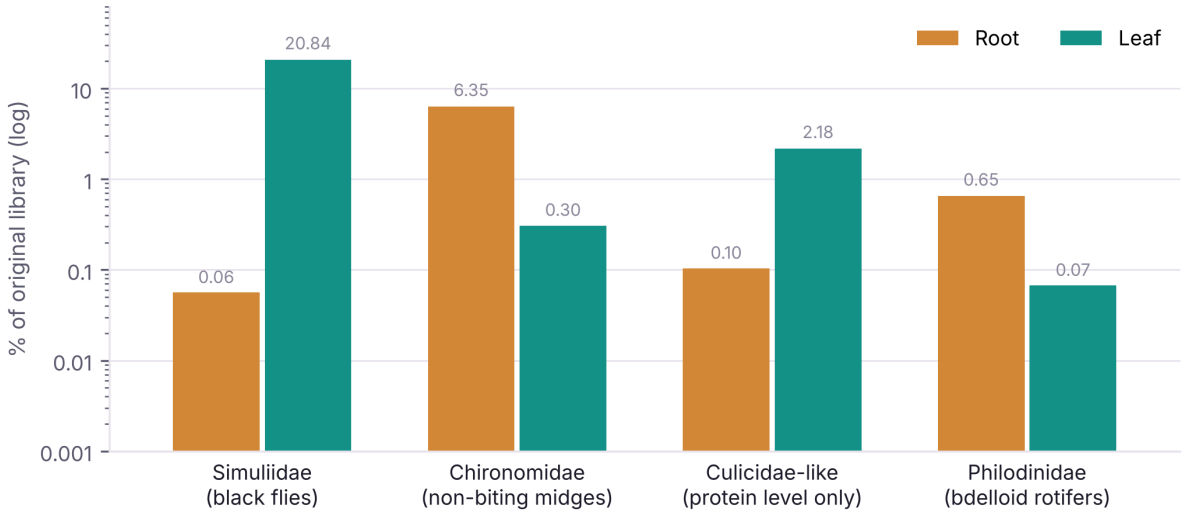
Taxon	Root	Leaf
Simuliidae (black flies)	0.06	20.84
Chironomidae (non-biting midges)	6.35	0.30
Culicidae-like (not confirmed by nucleotide evidence)	0.10	2.18
Philodinidae (bdelloid rotifers)	0.65	0.07

- *P. ceratophyllum* is a foundation species of eastern North American rivers and provides extensive habitat for invertebrates and fish [1, 2].
- The Chironomidae share excludes the fragments of the host nuclear rDNA isoform in gene cluster DN2745.
- These shares describe only the sources of library RNA; they say nothing about genus or species, numbers of individuals or relationships with the plant.

Main animal lineages

Family level, mean of three replicates

Chironomidae shown after removing a host nuclear 28S rDNA isoform



Source Report 4.5.2, Tables 71–72; 4.5.8, Table 94

Supplement S3

Bacteria: the two largest “bacterial” gene clusters are host plastid rRNA; the remaining bacterial signal is low

Method All isoforms vs host plastid, mitochondrial and rDNA references (host if a ≥ 97% segment covers ≥ half the isoform); per-isoform Salmon fragments in each library.

> About 40% of the original bacterial fragments over root and leaf (root about 20%, leaf about 78%) came from two rRNA-like gene clusters, which split by isoform into more than 98% host plastid rRNA. After deduction, bacteria are about 0.51% of the original library in root and 0.07% in leaf. The remaining bacterial signal is low: about 40% has no assigned phylum, the sequences re-verified to family or genus are mainly cyanobacteria and β- and α-proteobacteria, and the main taxa meeting the tissue-association criterion are root-biased.

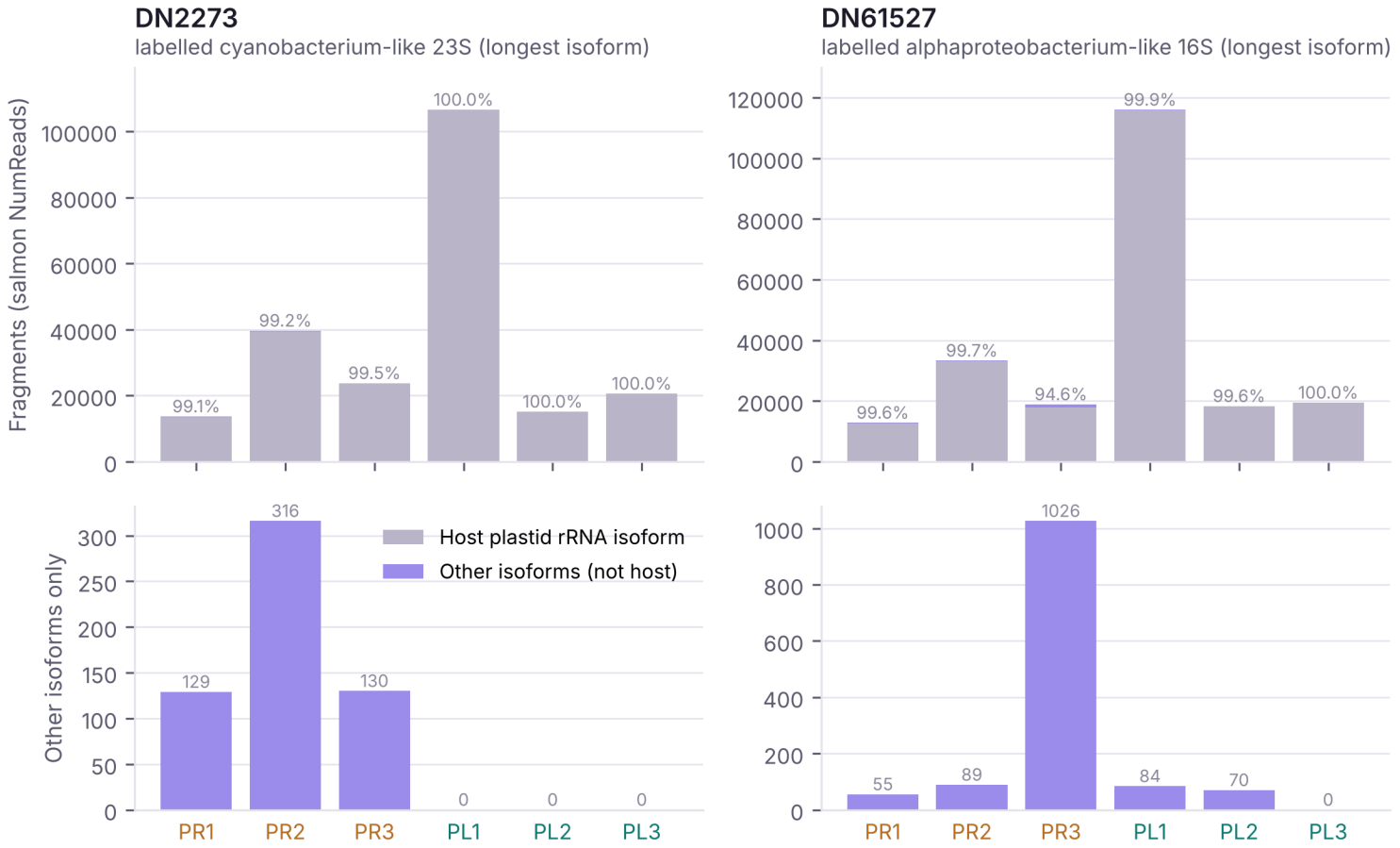
86 bacterial genes were re-verified, covering 58% of the root + leaf fragments of this major group before deduction; rank proportions below use only the re-verified part as denominator. The two clusters with the most fragments:

Gene cluster	Host plastid isoform	Share of gene fragments
DN61527	i9 (471 bp), 100% identity over full length	root 98.2%, leaf 99.9%
DN2273	i2 (476 bp), 99.2% identity, 99% coverage	root 99.3%, leaf 100%

Four lines of evidence for the two clusters, and the cause: slide 10.

Two 'bacterial' gene clusters are mostly host plastid rRNA

Fragments per root (PR) and leaf (PL) library on the host plastid rRNA isoform and on the other isoforms; labels on the top row are % host. Bottom row: the other isoforms alone, on their own scale; only the longest isoform of each cluster was searched against nt



Supplement S3

Bacteria: evidence and cause for the two mixed gene clusters

Method nt and NCBI 16S / LSU rRNA databases (megablast, dc-megablast) and host references; per-isoform Salmon fragments.

DN61527

Revision table: Methylobacteriaceae (protein level of script 84 only to Pseudomonadota) · gene level root 0.059%, leaf 0.134%

- **Evidence**: nt best hit *M. radiotolerans* 16S, 94.2%, full length; 16S-specific database best hit *Rubellimicrobium roseum*, 98.7%, but covering only 68%; host mitochondrion best 79%
- **Split**: isoform i9 (471 bp) is 100% identical to the host plastid over its full length and carries root 98.2% and leaf 99.9% of the gene's fragments
- **Handling**: fragments of the host plastid 16S rRNA isoform are deducted from the bacterial share. The other 6 isoforms are not host and have very few fragments: root libraries 55 / 89 / 1,026 (the 1,021 in PR3 from i1), leaf 84 / 70 / 0; root mean about 0.001%, single-library maximum about 0.003%. Only the longest, i2, was aligned (alphaproteobacterial 16S-like; the two databases disagree at order, no genus), and it has 17 fragments, all in PL1

DN2273

Original label Chamaesiphonaceae (*Chamaesiphon*-like, nt 23S 89.7%) · gene level root 0.070%, leaf 0.124%

- **Evidence**: nt best hit *Chamaesiphon minutus* 23S, 89.7%, full length; LSU-specific database best hit *Geitlerinema*, 83.9%; host mitochondrion best 87.3%
- **Split**: isoform i2 (476 bp) has 99.2% identity and 99% coverage to the host plastid and carries root 99.3% and leaf 100% of the gene's fragments
- **Handling**: fragments of the host plastid 23S rRNA isoform are deducted from the bacterial share. The longest non-host isoform (cyanobacterial 23S-like) reaches order (Gomontiellales) by nt, with fragments below 0.001% of the original library

Why whole clusters were counted as bacteria

- At gene level both are higher in leaf than in root, consistent in direction with host plastid rRNA being more abundant in green parts. Script 84 set gene labels from the longest transcript, which here was the bacteria-like isoform, so the fragments of the whole cluster were counted as bacteria; the host organelle rule of script 88 also aligned only the longest transcript and missed this.
- Host plastid 16S / 23S rRNA and bacterial rRNA share conserved regions (plastids derive from cyanobacteria), so Trinity grouped fragments of both into one gene cluster.
- In the scan of all 275,497 isoforms of non-plant genes, only these two and one trace sequence (DN231461, 0.001%) among bacteria contain host isoforms.

Source Report 4.5.1, Table 69; 4.5.8, Tables 92, 94

Supplement S3

Bacteria: re-verification coverage and reportable low-abundance taxa

Method nt, NCBI rRNA databases and NR (DIAMOND) for each gene; common ancestor of hits within 1%, environmental sequences removed; replicate CV and root/leaf ratio.

Coverage and reassignment

- After host-isoform fragments are deducted, the 86 re-verified genes cover 29.6% of the remaining bacterial fragments; within the re-verified part, ranks are major group 31%, family 26%, order 24% and genus 15%, mostly from rRNA-like sequences. The listed ranks add up to about 96%; the rest has no listed rank.
- Reassigned in re-verification: 3 bacterial labels, of which 1 rRNA-like sequence became red alga (Florideophyceae, 99.7%), 1 reached only other eukaryotes and 1 became a fungus (*Malassezia*); in addition, rRNA-like sequences with 2 stramenopile labels and 1 red-alga label became Cyanophyceae (85–89%).

Replicate stability

- By the criteria on slide 24, Oscillatoriaceae and Sphaerotilaceae are tissue-associated (root-biased). Microcoleaceae and 10 other low-abundance bacterial families (including Staphylococcaceae, treated as a common contaminant; 10 without it) are exploratory only and all root-biased; Sphingomonadaceae and Paracoccaceae are unstable across root replicates.

Bacteria assignable to family or genus

% of original library

Taxon	Entries	Root	Leaf
Oscillatoriaceae, *Phormidium* candidate	1	0.0149	0.0004
Oscillatoriaceae, genus undetermined	2	0.0070	0.0001
Both Oscillatoriaceae rows		~0.0219	0.0005
Sphaerotilaceae	5	0.0109	0.0009
Hapalosiphonaceae	1	0.0054	0.0002

- The other main taxa listed are below about 0.006% in root. Low-share signals of *Staphylococcus* and *Pseudomonas* are of common environmental or human origin and are not listed as stable associated taxa.
- One Nitrospiraceae sequence was re-verified to family, with near-zero fragments; its functional clue is on slide 15.

Source Report 4.5.8, Tables 91, 93; Appendix B, Table 102

Supplement S4

Fungi: chytrid-related signal root-biased and stable; other candidates reported by rank

Method Same re-verification with 18S / 28S / ITS databases; family threshold protein ≥ 80% or nucleotide ≥ 90%; replicate CV within root and within leaf.

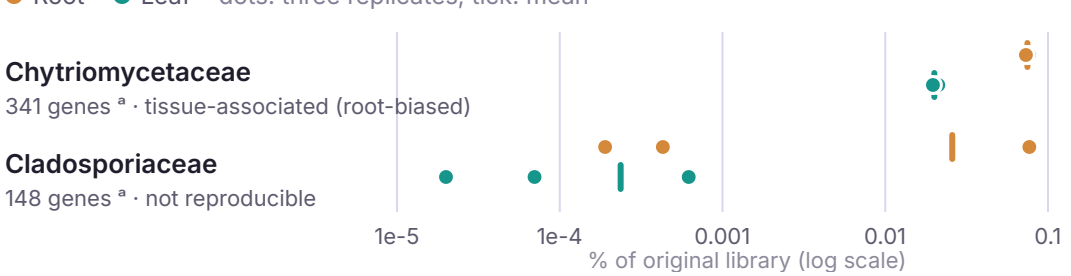
> The chytrid-related signal that reaches the family threshold is higher in root and stable across replicates within root and within leaf; in the re-verified part, about half of the fungal fragments support only the major group Fungi.

60 fungal genes were re-verified, covering 32.5% of the group's root + leaf fragments. Within them: major group 50%, family 16%, genus 11%, class 9%, order 6%, and about 8% moved to other eukaryotes or stramenopiles (re-verified part only, not all fungal fragments).

Chytrid-related Chytriomycetaceae: two definitions (% of original library)

Definition	Genes [a]	Root	Leaf
Sum over revision-table (V1) labels (script 87)	2,782	0.261	0.068
Evidence at the family threshold (protein ≥ 80% or nucleotide ≥ 90%)	341	0.075	0.020

[a] Both rows count all nine sequenced libraries, whole plant included. Row 1: genes with ≥ 10 fragments in any library; row 2: the most genes with ≥ 10 fragments in any one library (stability criterion). At the family threshold, replicate CV is 0.02 in root and 0.05 in leaf: tissue-associated (root-biased). None of the 23 re-verified Chytriomycetaceae-labelled sequences reached family (major group 14, class 6, order 1; 2 to other eukaryotes); the 3 that reach Chytriomycetaceae by nt had original labels Fungi (no family), Lobulomycetaceae and Aspergillaceae.



Other fungi assignable to family or genus (% of original library)

Taxon	Entries	Root / leaf	Note
Pleomassariaceae	1	0.035 / 0.000	LSU-specific database
Cladosporium	2	0.023 / 0.000	rRNA-specific databases; summed genus-level labels have root replicate CV 1.71, not reproducible, with the signal from a single replicate
Williopsis (Wickerhamomycetaceae)	1	0.017 / 0.000	
Pleosporaceae	1	0.014 / 0.000	Root-biased, but unstable across root replicates

- In read tracking, one fungus-like source has the closest protein reference *Sordaria macrospora* (78.9%), not enough to identify a species.
- Fungal labels moved to other sources: the rotifer (Philodinidae) rRNA and the grass sequence (script 86); 2 Chytriomycetaceae-labelled sequences (protein 78.7%, nt 76.4%) reach only Eukaryota; 1 rRNA-like sequence is a stramenopile by the eukaryotic LSU-specific database at 89.1% (the raphidophyte *Chattonella*), 1 is Bacillariophyceae by nt at 91.9%, and several more reach only Eukaryota (script 88). Fungi in root: 1.225% → 0.887%.

Source Report 4.5.1, Table 69; 4.5.3, Table 76; 4.5.8, Tables 91, 93; Appendix B, Table 102

Supplement S5

Algae: diatoms have low-share genus candidates; most re-verified red and green algal fragments reach only broad groups

Method Same re-verification with SSU / LSU databases and NR; identity-capped ranks; replicate CV and root/leaf ratio per family and genus.

> Some diatom-related families and genera are stable across root replicates and root-biased; most re-verified red and green algal fragments reach only the major group or phylum, with a few family or genus candidates. The three groups together make up about 2.23% of the root original library.

Group	Re-verification coverage and ranks	Reportable taxa (% of original library, root / leaf)	Replicate stability
Diatoms and other stramenopiles root 1.78%, leaf 0.27%	60 genes cover 22.5% of fragments: major group 70%, order 9%, family 8%, phylum 7%, class 4%, genus 2%	Fragilariaceae: 4 entries to family (0.014 / 0.013), 3 to *Fragilaria* (protein ≥ 90%); *Gomphonema* (SSU-specific database), *Fragilariopsis*, Naviculaceae, Bacillariaceae and Plagiogrammaceae 1 each	Under re-verified labels, Fragilariaceae (up to 723 genes in one library; 0.074 / 0.025), Naviculaceae (814; 0.090 / 0.004), Bacillariaceae and Thalassiosiraceae are all tissue-associated (root-biased); at genus level *Fragilaria* (0.025 / 0.004) and *Seminavis* (0.020 / 0.001) are also root-biased
Diatom-like	*Melosira dubia* 28S, identity 80.4%, coverage 70%	Below the order threshold (85%); recorded as a diatom-like sequence, family undetermined	—
Red algae root 0.40%, leaf 0.11%	60 genes cover 39.8% of fragments (after deducting host isoforms): major group 84%, phylum 7%, class 7%, genus 2%	The former largest entry (original stramenopile label, closest reference *Palmaria* 93%, root 0.09%, leaf 0.11%) has 97–99.9% of its fragments on host plastid rRNA isoforms and is recorded as host; its remaining red-alga-like isoforms reach only Florideophyceae, with very few fragments. *Acrochaetium* and *Audouinella* 1 each to genus (0.001 / 0.000). DN7210, an 18S sequence (closest reference *Audouinella*, 90.4%), was reassigned to Florideophyceae on 2026-10-28 (0.076 / 0.067); it is in the totals on the left but not among the 60 re-verified genes	Acrochaetiaceae: low abundance, root-biased, exploratory only
Green algae etc. root 0.05%, leaf 0.06%	60 genes cover 47.6% of fragments: phylum 49%, major group 36%, order 4%; 6% reassigned to embryophytes, 4% to other eukaryotes	*Cladophora* 1 (protein) and Dunaliellaceae 1, near-zero fragments; other re-verified fragments mainly reach only phylum or major group	No taxon entered the assessment

Definitions: coverage is relative to all root + leaf fragments of the major group, rank percentages to the re-verified part. Family and genus candidates are the rank currently reportable, not species identification.

Source Report 4.5.1, Table 69; 4.5.3, Table 76; 4.5.8, Tables 91, 93

Supplement S5

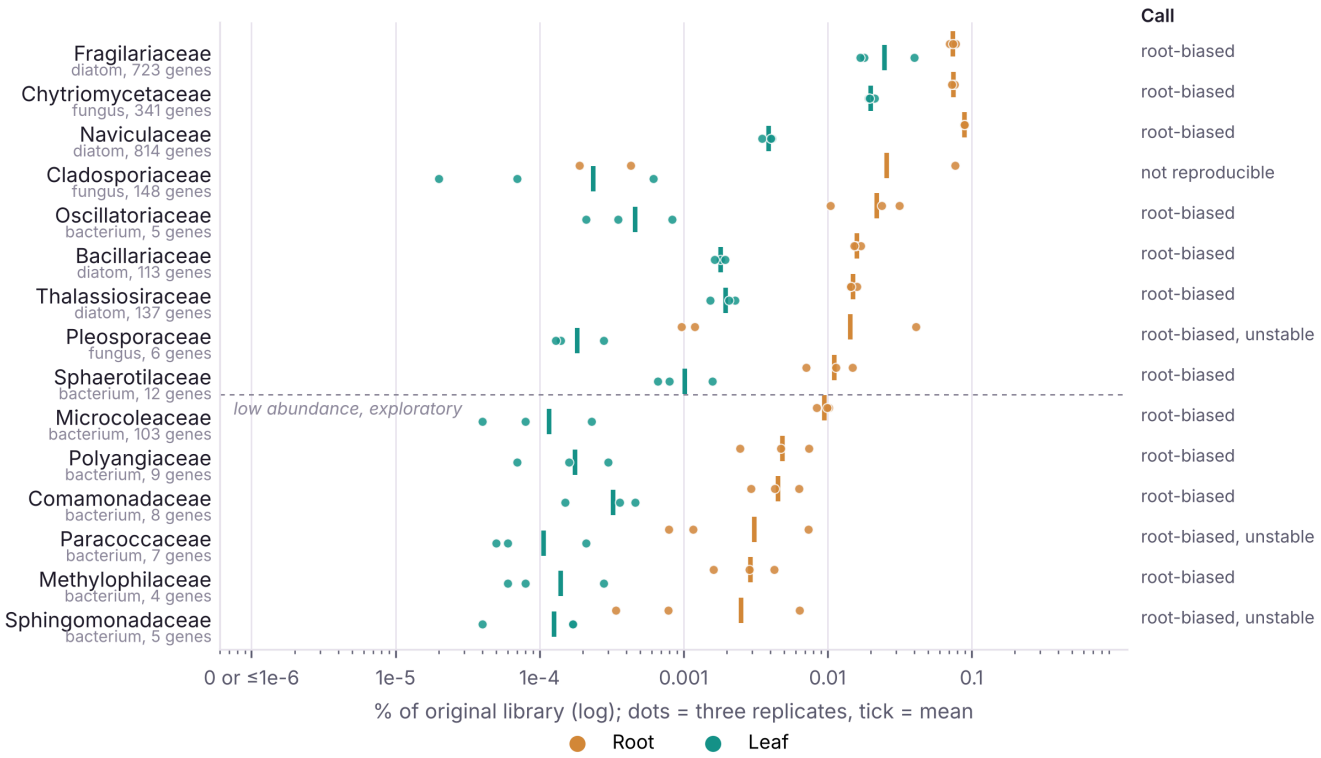
Algae: hit identities and reassignments

Method Protein (DIAMOND vs NR) and nucleotide (megablast vs nt, 18S database) identities; common ancestor with and without environmental sequences.

- The median protein-hit identity of all sequences under the Naviculaceae and Fragilariaceae labels is only 66% and 62%. At family level, the “re-verified labels” on slide 13 keep only sequences whose evidence reaches the family threshold (protein ≥ 80% or nucleotide ≥ 90%), so family totals are far smaller than under the original labels.
- One *Fragilaria crotonensis* protein hit is only 58.7% and is not used for species identification.
- DN7210: the best hit is the red alga *Audouinella* 18S (90.4%, full length). Of the 10 hits within 1 percentage point, 9 are red algae and 1 is an environmental sequence without taxonomic information; the common ancestor computed by script 86 included it and reached only Eukaryota. Without the environmental sequence the common ancestor is Florideophyceae, and the sequence was reassigned to red algae (Florideophyceae) on 2026-10-28; identity is low, so no family or genus is assigned.
- Reassigned in re-verification: rRNA-like sequences with 2 stramenopile labels and 1 red-alga label became Cyanophyceae; 3 sequences labelled green algae are embryophytes by nt (75.5%–85.3%) and were removed from non-plant.

Microbial and algal families after host isoform removal

Top: the 9 families entering the stability test (mean ≥ 0.01% in root or leaf; ≥ 3 genes with ≥ 10 fragments in one library)
Below the dashed line: 6 of the 12 families at 0.002–0.01%, exploratory only (Staphylococcaceae, a common contaminant, not shown)
Gene counts under the names: the largest number of genes with ≥ 10 fragments in any single library (all nine sequenced libraries counted)



Left, fig19_families: bacterial, fungal and algal families in the root and leaf libraries (dots for the three replicates, tick for the mean), in descending order of the sum of root and leaf means.

Source Report 4.4.10, Table 64; 4.5.1, Table 69; 4.5.2, Table 71; Appendix B, Table 102

Supplement S6

Bacterial functional clues: detected, but no complete pathway

Method eggNOG-mapper 2.1.12 KO assignments mapped to KEGG modules, plus NR best-hit names; bacterial and archaeal genes only.

After re-verification, 29 prokaryotic function-candidate records remain, from 27 distinct genes (2 genes fall in two functional categories each). Each pathway has only single subunits or scattered steps, which do not support a complete metabolic process.

Functional clue	Entries	Components from NR best hits	Matching components not detected, and limits
Nitrification M00528 / M00804	3	amoC 2 (identity 100%; one from Nitrospiraceae); 1 monooxygenase-subunit-like (72%; ammonia and methane cannot be separated)	amoA, amoB, hao, nxrA and nxrB all not detected
Methane or methanol oxidation M00174	4	PQQ-dependent methanol / ethanol dehydrogenase-like 2 (94–95%; mxaF / xoxF cannot be separated from ethanol dehydrogenase); the other 2 are the nitrification row's monooxygenase-subunit-like and Nitrospiraceae amoC-like sequences (script 84 names include ammonia and methane monooxygenase)	pmoA, pmoB, sMMO and mxaI not detected; both the monooxygenases and the dehydrogenases are functionally ambiguous
Nitrogen fixation M00175	3	nifH-like (dinitrogenase reductase) 1, 100% to an uncultured bacterium; C-terminal fragment of Mo nitrogenase 1 (69%); nitrogenase-like 1 (72%, closest reference a cyanobacterium)	nifD, nifK and V nitrogenase not detected
B12 synthesis M00122, M00924, M00925	19	By KO only cobO 1; by NR name also cobQ 2 and 1 each of cobS, cobD / cbiB, cbiX, cobN, cobM, cobL, cobG, cobA, cysG and sirohydrochlorin chelatase, from different groups such as *Pseudomonas*, cyanobacteria and uncultured bacteria	The three modules have 1 / 0 / 0 components by KO and 4 / 2 / 5 by NR name; 5 others are not synthesis steps (CobW-type, Sir2-type, BolA, cobalt transporter, archaeal cyclic 2,3-diphosphoglycerate synthase); 1 aminotransferase (eggNOG name cobD) undecided

3 / 4 / 3 / 19 are entries per category (29 records); by gene they are 27, because the two monooxygenase sequences (subunit-like and Nitrospiraceae amoC-like) count under both nitrification and methane oxidation. Taxonomic and functional evidence are listed separately.

FRAGMENTS AND TWO DEFINITIONS

Each candidate is below 0.0001% of the original library; most KOs are empty, and across seven modules KO finds only cobO. NR names are the broad definition; the two are counted separately.

AMBIGUITY FROM HOMOLOGY

Ammonia and methane monooxygenases are homologous (old KEGG K10944–K10946 do not separate them); nifH is homologous to protochlorophyllide reductase ChlL / BchL; cysG / cobA also make siroheme; CobW-type G3E GTPases are not B12-specific. Every candidate is flagged as ambiguous.

EUKARYOTIC NAME MATCHES

Keyword matches to eukaryotic genes with the same names are listed separately, not used as functional evidence.

SCOPE OF CONCLUSIONS

Single subunits or a few steps show only that related gene fragments were detected, not a complete process; Poly(A) selection misses most bacterial mRNA.

Source Report 4.5.4, Tables 78–79

Supplement S7

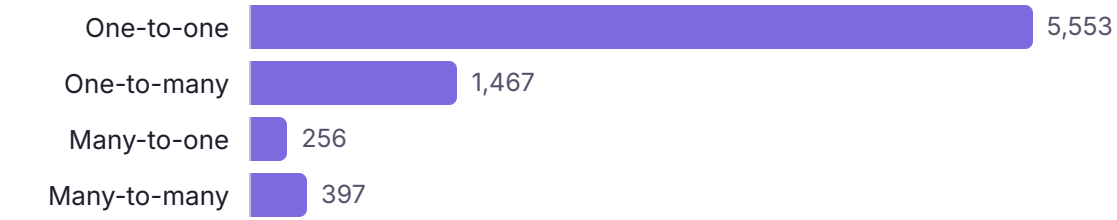
Proteome-wide correspondence with *Arabidopsis*

Method Proteinortho 6.3.6 (DIAMOND; E ≤ 1e-5, identity ≥ 25%, coverage ≥ 50%); cross-check with eggNOG-mapper orthologous groups.

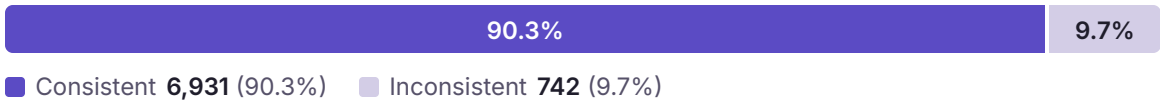
> Many *P. ceratophyllum* proteins have *Arabidopsis* counterparts; one-to-one correspondences come from a proteome-wide first pass, and not all have been confirmed by gene trees.

- 7,327 orthogroups contain both *P. ceratophyllum* and *Arabidopsis*, involving 7,673 proteins of *P. ceratophyllum* [4].
- One-to-many and many-to-many correspondences keep joint labels rather than forcing a single *Arabidopsis* gene.
- The 5,553 one-to-one pairs come from Proteinortho and were not confirmed one by one with gene trees [4].
- eggNOG orthologous-group check: 6,931 consistent and 742 inconsistent (about 90.3% consistent) [5].

Entries by correspondence type (*P. ceratophyllum* to *Arabidopsis*)



eggNOG orthologous-group check



Source Report 4.3.3, Table 29

Supplement S8

What the key vegetative developmental markers represent

The 24 markers fall into four modules (shoot apex, leaf, root, vascular), each standing for a developmental function known in *Arabidopsis*.

Shoot apical meristem

- **STM** *SHOOT MERISTEMLESS*: class I KNOX homeobox transcription factor, required to form and maintain the shoot apical meristem [6]
- **KNAT1** *KNOTTED-LIKE FROM ARABIDOPSIS THALIANA 1 (BREVIPEDICELLUS)*: class I KNOX; affects inflorescence internode and pedicel development [7]
- **KNAT2, KNAT6** *KNOTTED-LIKE 2, 6*: class I KNOX, like STM and KNAT1
- **WUS** *WUSCHEL*: homeobox transcription factor of the organizing center; keeps the cells above it as stem cells [8]
- **CLV3** *CLAVATA3*: small peptide signal secreted by stem cells; limits the size of the shoot apical meristem [9]

Leaf

- **AS1** *ASYMMETRIC LEAVES1*: MYB-type (ARP) transcription factor; represses KNOX genes in leaf primordia and establishes leaf form [10, 11]
- **FIL, YAB2, YAB3, YAB5** *FILAMENTOUS FLOWER (YAB1)* and YABBY family members: FIL, YAB2 and YAB3 specify abaxial (lower-side) identity of lateral organs [12]; YAB5 belongs to the same family
- **CUC1, CUC2, CUC3** *CUP-SHAPED COTYLEDON 1–3*: NAC transcription factors for boundaries between organs and for establishing the shoot apical meristem; CUC3 has been studied specifically in the specialized shoots of Podostemaceae [13, 14]

Root

- **SHR** *SHORT-ROOT*: GRAS transcription factor expressed in the stele; through radial signaling it determines endodermis and cortex differentiation [15]
- **SCR** *SCARECROW*: GRAS transcription factor controlling the asymmetric division that produces endodermis and cortex [16]; also involved in bundle sheath development in leaves [17]
- **PLT1, PLT2** *PLETHORA 1, 2*: AP2-type transcription factors; establish the root stem cell niche [18]
- **WOX5** *WUSCHEL-RELATED HOMEOBOX 5*: expressed in the root quiescent center; maintains the surrounding stem cells [19]

Vascular

- **PXY** *PHLOEM INTERCALATED WITH XYLEM (TDR)*: receptor kinase on cambium cells; receives the TDIF signal and maintains the arrangement of vascular tissues [20, 21]
- **CLE41, CLE44** CLE peptides (TDIF): phloem-derived peptides that suppress differentiation of vascular stem cells into xylem [21, 22]
- **WOX4** *WUSCHEL-RELATED HOMEOBOX 4*: downstream of TDIF–PXY signaling; promotes proliferation of cambium cells [23]
- **HB8** *ARABIDOPSIS THALIANA HOMEOBOX 8 (ATHB-8)*: class III HD-ZIP; procambium and vascular differentiation [24]

In Podostemaceae (Old World species), the expression sites of STM, WUS and AS1 homologs in shoots have been studied by in situ hybridization [25]; this study has only whole-organ transcriptomes and asks how much these homologs are expressed in root and leaf.

Source Report 4.3.6, Table 35; references [6–25]

Supplement S8

Key vegetative markers by level of homology evidence

Method Pfam domain recall (HMMER 3.4) → MAFFT L-INS-i → IQ-TREE 2.4.0 (ModelFinder; 1,000 UFBoot and SH-aLRT) → species-overlap orthology; tblastn re-query for misses.

> Key markers have been rechecked with gene trees [26, 27] and against the pre-screening assembly; a few relationships remain undecided.

Only the 24 vegetative development targets of *Arabidopsis* are counted, not floral organ markers; numbers in the table are *Arabidopsis* targets, not genes of *P. ceratophyllum*. Sequences and evidence for each target: slide 33 (Appendix B).

Module	High confidence 8	Medium confidence 7	Undecided 3	Not detected this time 4	TDIF-class motif evidence 2
Shoot apical meristem	WUS	STM, KNAT1, KNAT2, KNAT6		CLV3	
Leaf	AS1, YAB5, CUC3	YAB3, CUC2	FIL	YAB2, CUC1	
Root	SHR, SCR		PLT1, PLT2	WOX5	
Vascular	WOX4, PXY	HB8			CLE41, CLE44

- KNAT2/6 correspond to one sequence; the joint label is kept.
- HB8 carries the ATHB-15/HB8 joint label in the formal-assembly tree.
- YAB3 has one-sided support only (the seven-species tree of the pre-screening assembly), so it is at most medium.
- CLE41/44 have only two TDIF-class candidates, which cannot be assigned individually.

Source Report 4.3.5–4.3.6, Table 35

Supplement S10

Overall expression differences between root and leaf

Method Salmon 1.10.3 → tximport → DESeq2 1.50.2 (~ tissue; BH padj < 0.05 and |log2FC| ≥ 1); GO by topGO, KEGG by hypergeometric test.

> Photosynthesis-related signals stand out in leaf; root is enriched for ribosome- and translation-related transcripts.

HIGHER IN LEAF

2,461 genes

Enriched for light reactions, antenna proteins, porphyrin metabolism and the Calvin cycle

HIGHER IN ROOT

2,608 genes

Enriched for cytosolic ribosome, translation and transcriptional regulation terms

- In the strict main analysis, 14,010 genes entered testing [28, 29]: 2,608 are higher in root and 2,461 higher in leaf, 5,069 in total.
- The 5,069 genes differ significantly from zero with a fold-change point estimate of at least two; in the threshold test with H0 |log2FC| ≤ 1 and H1 |log2FC| > 1, 2,466 are significant. Both definitions are reported.
- The current model is ~ tissue, without same-plant pairing. Once the number correspondence and the independent sampling units are confirmed, the test will follow the actual pairing, and the calls on slides 20–22 and 27–28 will be rechecked.

Source Report 4.4.2, Tables 37–38; 4.4.3, Tables 40–41

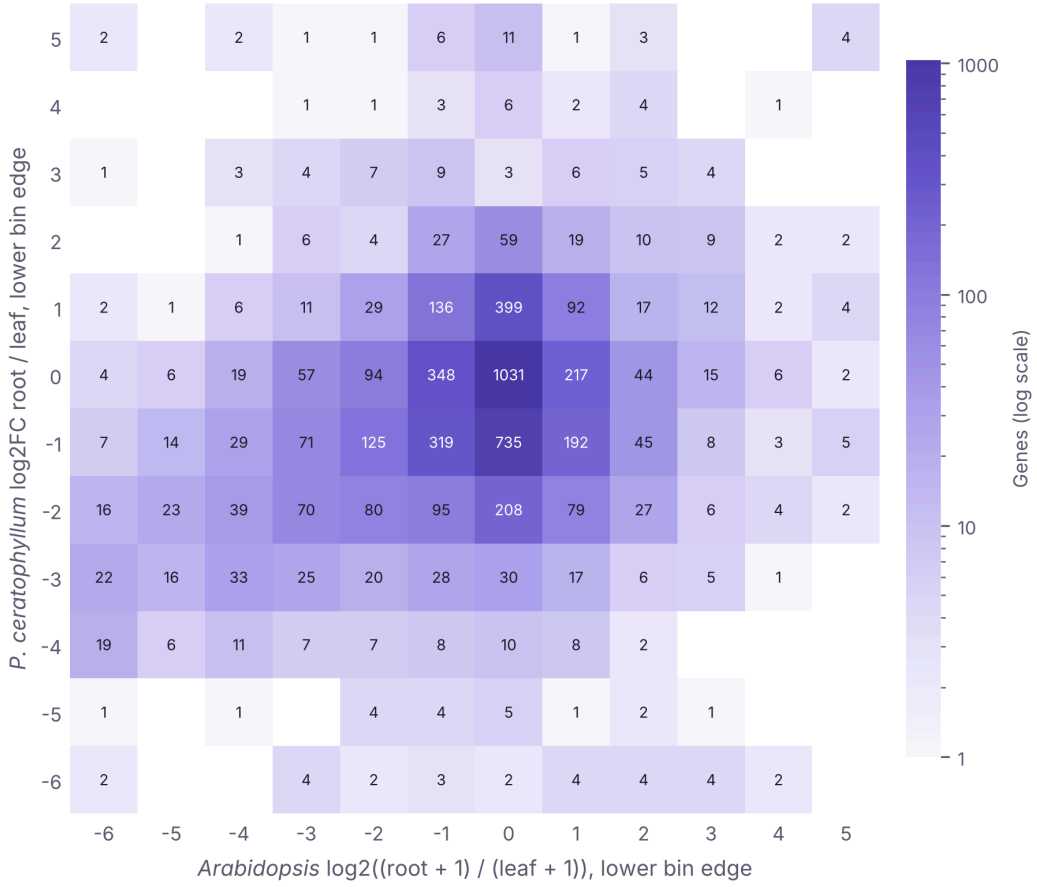
Supplement S10

Root–leaf differences in the two species

5,313 one-to-one genes with values in both species; Spearman 0.18

Arabidopsis: E-MTAB-7978 tissue atlas; Spearman 0.19 for the 4,755 genes with root or leaf ≥ 5 TPM (script 82)

1-unit bins labelled by lower edge; on both axes the first and last bins also hold values beyond ±6

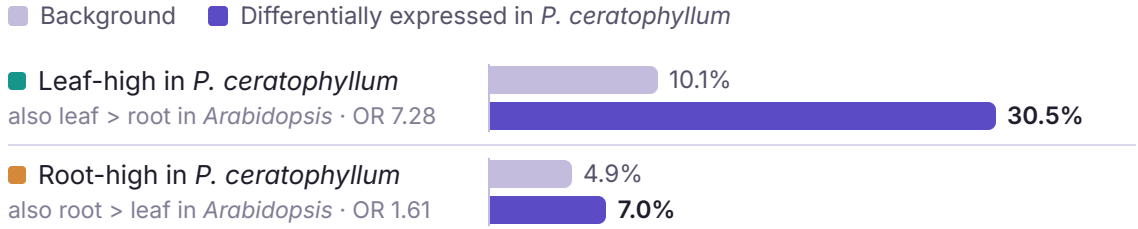


Compared with *Arabidopsis*, directions agree more strongly on the leaf side

Method *Arabidopsis* organ TPM from Expression Atlas E-MTAB-7978; one-sided Fisher tests and Spearman correlation on one-to-one orthologues.

> Across the two species, root–leaf difference directions agree overall: strongly on the leaf side, weakly on the root side.

5,392 genes entered the overall one-to-one comparison; *Arabidopsis* organ expression is from the Expression Atlas tissue atlas [30].



- Of the genes higher in leaf in *P. ceratophyllum*, 30.5% are also higher in leaf than in root in *Arabidopsis* (background 10.1%, OR 7.28).
- Of the genes higher in root, 7.0% are also higher in root than in leaf in *Arabidopsis* (background 4.9%, OR 1.61).
- The Spearman correlation of root–leaf differences between the two species is 0.192 (4,755 one-to-one genes).

Left: 5,313 one-to-one genes without an atlas TPM filter give Spearman 0.18; the 4,755 genes passing the filter of script 82 give 0.19.

Source Report 4.4.7, Table 52; Appendix B, Table 106

Supplement S9

STM is higher in root; several developmental markers are expressed in both root and leaf

Method Gene-level TPM (Salmon + tximport); strict DESeq2 root vs leaf test; orthology from focused gene trees.

> There are clear developmental candidate signals, but expression in bulk tissue cannot show cellular location.

Marker	Represents	Root TPM	Leaf TPM	Current result
STM	shoot meristem maintenance (KNOX)	3.09	0.37	● **Root-high**, padj 0.0021; homology medium
KNAT1	class I KNOX	21.86	19.77	Not significant; homology medium
AS1	represses KNOX in leaves (ARP)	39.79	28.84	Not significant; homology high
SCR	endodermis / bundle sheath (GRAS)	23.87	44.78	● **Leaf-high**; homology high
SHR	endodermis differentiation (GRAS)	1.20	1.23	Not significant
WUS	shoot stem cell maintenance	0.40	0.60	Low; the sequence covers only part of the protein

- Higher STM in root bears on the research question of shoots arising from roots; where it is expressed cannot be judged from bulk material [25, 31].
- In *Arabidopsis*, SCR also acts in bundle sheath development in leaves [17], so higher expression in leaf alone is not evidence of changed organ identity.
- WOX5 not detected and PLT1/2 undecided do not mean that the genes are lost.
- PXY and HB8 are not significant, WOX4 is very low, and the TDIF-class candidates are not in the strict test; no evidence yet of a coordinated change in the vascular module.

Vegetative marker genes in root and leaf

Gene-level TPM in the three root and three leaf libraries; grey labels: orthology unresolved or not detected; empty rows: no sequence assigned. Root vs leaf: strict DESeq2 call; low = too few reads to test; nohit = not in the strict plant gene set. Orthology: confidence of the *P. ceratophyllum* orthologue assignment



Source Report 4.4.6, Tables 48–49; Appendix B, Table 103

Supplement S10

Extended candidates: concordant and opposite directions

Method Strict DE genes × one-to-one or tree orthologues × *Arabidopsis* organ-specific genes (≥ 4-fold); robustness under four atlas treatments.

The overall differences have been narrowed down to specific candidates, and the robustness of the atlas categories was checked.

85	14	11	60
main-list candidates	same direction	opposite direction: all leaf-high here and root-specific in *Arabidopsis*	correspond to other *Arabidopsis* organs

Candidate	Represents	Root / leaf TPM	Root Leaf (log)	Main point
LBD14	LOB-domain transcription factor	84.71 / 13.33		Root-high; atlas category robust
TIP4-1	tonoplast aquaporin	59.57 / 117.67		Leaf-high; root-specific in *Arabidopsis*
SAUR55	early auxin-responsive small protein (SAUR family)	3.62 / 57.17		Leaf-high; root-specific in *Arabidopsis*; the Swiss-Prot best hit is another member, SAUR12, to be confirmed by a family tree
CHIB1	acidic endochitinase	0.39 / 13.53		Leaf-high; root-specific in *Arabidopsis*
ADH1	alcohol dehydrogenase, a common hypoxia-response gene	213.80 / 10.60		About 17-fold higher in root (DESeq2; TPM ratio about 20); *Arabidopsis* category unstable, read only as a difference within *P. ceratophyllum*

- Of the 85, 51 keep the same category under four atlas treatments, and 33 still do when the threshold is raised to 8-fold.
- ADH1, BLH8 and PER21 have unstable *Arabidopsis* categories; their root–leaf differences in *P. ceratophyllum* can still be reported.

Source Report 4.4.8, Tables 53, 55–57; 4.4.9, Tables 59–60

Identification ranks and replicate stability of bacteria, fungi and algae: sequence-by-sequence re-verification

Method 386 genes: nt and NCBI rRNA databases (megablast, dc-megablast), NR (DIAMOND); common ancestor of hits within 1%; ranks capped by identity.

> After taxonomic evidence was added for 386 bacterial, fungal and algal candidate genes, most re-verified fragments still support only broad groups. Of the 9 families in the main criterion, 8 are root-biased and 1 is not reproducible; no family or genus under the re-verified labels meets the predefined definition of "consistent across samples".

Major group	Re-verified genes; coverage of the group's fragments (root + leaf)	Reported ranks (fragment-weighted)	Taxa reportable to family / genus	Reassigned or removed
Bacteria	86; 29.6% after deducting host isoforms	major group 31%, family 26%, order 24%, genus 15% (before deduction, "genus 39%" and "order 42%" were mainly the host plastid rRNA of DN61527 and DN2273; slide 9)	Low-share family and genus candidates such as Oscillatoriaceae (*Phormidium*), Sphaerotilaceae, Hapalosiphonaceae, Polyangiaceae and Comamonadaceae; the two Oscillatoriaceae rows sum to about 0.0219% of root, and the genus-level row is not the family total	3 reassigned (1 each to red algae, other eukaryotes and fungi); host plastid rRNA isoforms of the two gene clusters removed
Archaea root 0.009%	60; 50%	major group 92%	None	8% to other eukaryotes or fungi
Fungi	60; 32.5%	major group 50%, family 16%, genus 11%, class 9%, order 6%	Pleomassariaceae, *Cladosporium*, *Williopsis*, Pleosporaceae, Chytriomycetaceae (3 entries)	8% to other eukaryotes or stramenopiles
Stramenopiles diatoms etc.	60; 22.5%	major group 70%, order 9%, family 8%, phylum 7%, class 4%, genus 2%	Fragilariaceae (*Fragilaria*), *Gomphonema*, *Fragilariopsis*, Bacillariaceae, Plagiogrammaceae, Naviculaceae	1% to cyanobacteria
Red algae	60; 39.8% after deduction	major group 84%, phylum 7%, class 7%, genus 2%	*Acrochaetium*, *Audouinella*	1% to cyanobacteria; host plastid rRNA isoforms of DN22134 removed; DN7210 moved in from other eukaryotes (Florideophyceae; not among the 60)
Green algae etc.	60; 48%	phylum 49%, major group 36%, order 4%, genus 1%, family 1%, class 1%	*Cladophora*, Dunaliellaceae	6% embryophytes (removed), 4% to other eukaryotes

Denominators: coverage is the share of the group's root + leaf fragments carried by the re-verified genes; rank distributions are fragment-weighted within the re-verified part and do not mean every sequence was ranked. **Approach**: high-fragment sequences first (10–60 genes per group; for bacteria, plus key functional and flagged sequences), adding nt, rRNA-specific databases [33], NR near-score hits [34] and host references; environmental-sample sequences are left out of ranking, and the descriptive rank thresholds are not species-identification standards.

Source Report 4.5.8, Tables 91, 93

Supplement S6

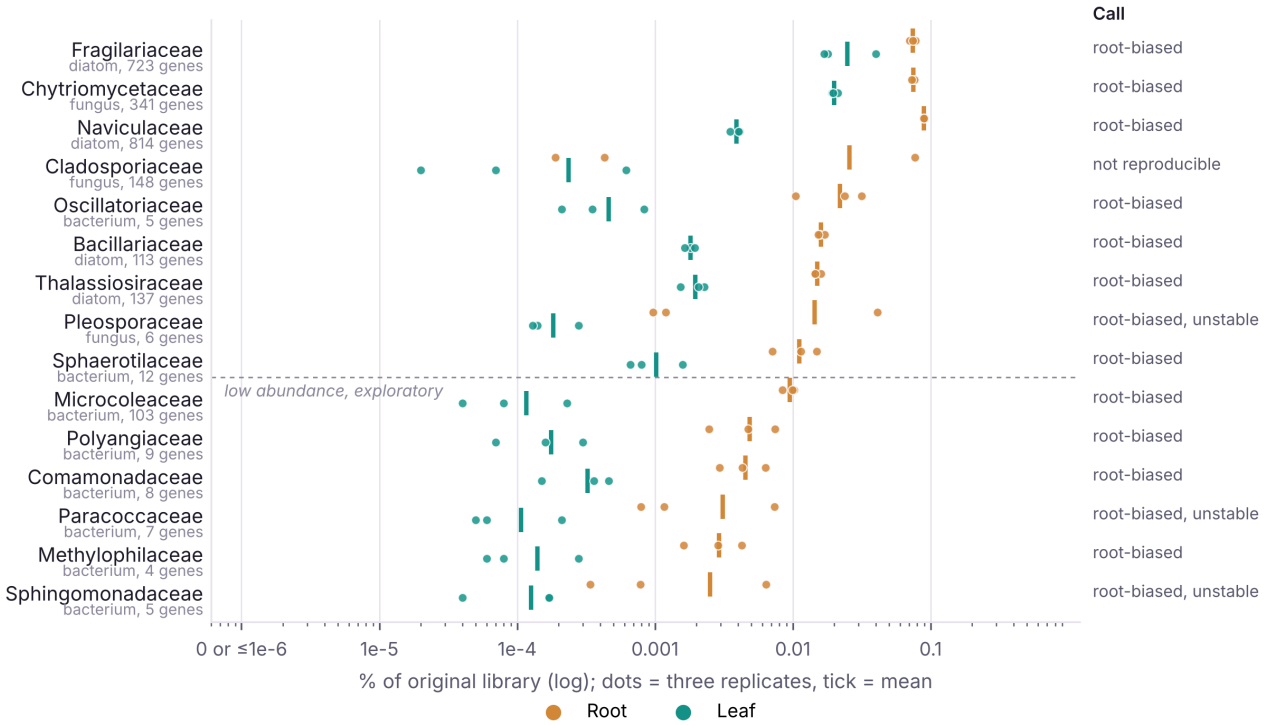
Replicate stability and compositional distance

Method Per family: entry at mean ≥ 0.01% and ≥ 3 genes; replicate CV and root/leaf ratio; Bray–Curtis distance with a permutation test.

- **Replicate stability** (criterion of script 84: root or leaf mean ≥ 0.01% and ≥ 3 genes detected; re-verified labels): 9 families entered the assessment. 8 are tissue-associated (root-biased; the root signal of Pleosporaceae comes almost only from PR3) and 1 is not reproducible (Cladosporiaceae); of 3 genera, *Fragilaria* and *Seminavis* are root-biased and *Cladosporium* is not reproducible. No family or genus meets the predefined condition "consistent across samples", which does not mean root and leaf cannot each be stable: Chytriomycetaceae has root and leaf CV of 0.02 and 0.05 but a higher level in root. Abundant enough but with too few genes, not assessed: Pleomassariaceae, Wickerhamomycetaceae.
- **Low abundance** (0.002–0.01%): the 12 families treated as exploratory only (11 bacterial, 1 red algal) are all root-biased, and Sphingomonadaceae and Paracoccaceae are unstable across root replicates. Before and after isoform-based deduction of host sequences, the lists and conclusions for the 9 assessed families and the 12 low-abundance families are unchanged; Roseobacteraceae, earlier not assessed for too few genes, falls below the entry threshold after deduction.
- **Compositional distance**: Bray–Curtis distance of the composition within bacteria, fungi and algae (family level, after deducting host isoforms). Root vs leaf: within-group 0.078 and between-group 0.173, permutation p = 0.10, not significant at the usual 0.05 level. The current unpaired, exchangeable 3 vs 3 design has a minimum p of 0.10; results are described with distances and replicate distributions, and the permutation design will be rechecked once the actual pairing and independence are confirmed.
- **Overall**: most families and genera meeting the criteria are root-biased. Taxonomic resolution is limited, and the re-verified part can mostly be reported only as broad groups, which is related to Poly(A) libraries capturing mainly partial rRNA fragments and conserved proteins. The root–leaf comparison alone is currently described with effect sizes and replicate distributions.

Microbial and algal families after host isoform removal

Top: the 9 families entering the stability test (mean ≥ 0.01% in root or leaf; ≥ 3 genes with ≥ 10 fragments in one library)
Below the dashed line: 6 of the 12 families at 0.002–0.01%, exploratory only (Staphylococcaceae, a common contaminant, not shown)
Gene counts under the names: the largest number of genes with ≥ 10 fragments in any single library (all nine sequenced libraries counted)



Source Report 4.5.3, Table 73; 4.5.8, Tables 93, 95; Appendix B, Table 102

Supplement S12

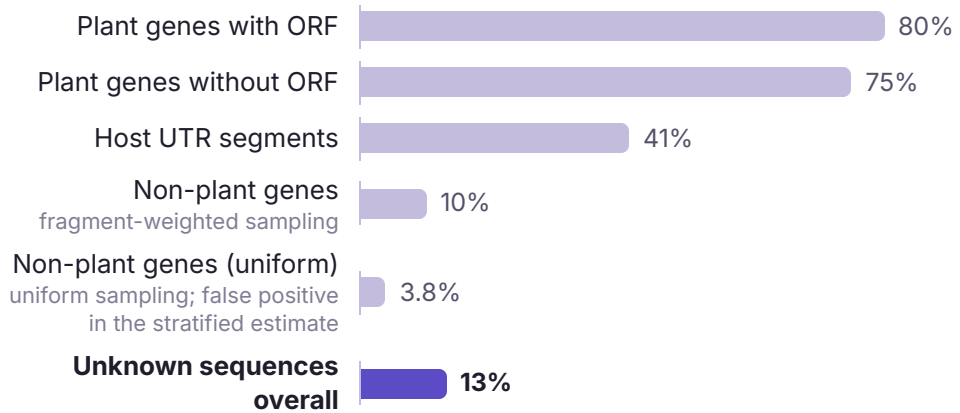
“Unknown” sequences: host or foreign?

Method dc-megablast to the *Cladopus* genome, stratified by ORF × length with sensitivity and false-positive controls; bootstrap intervals. Sampling frame and fragment weights pooled over all nine sequenced libraries (whole plant included).

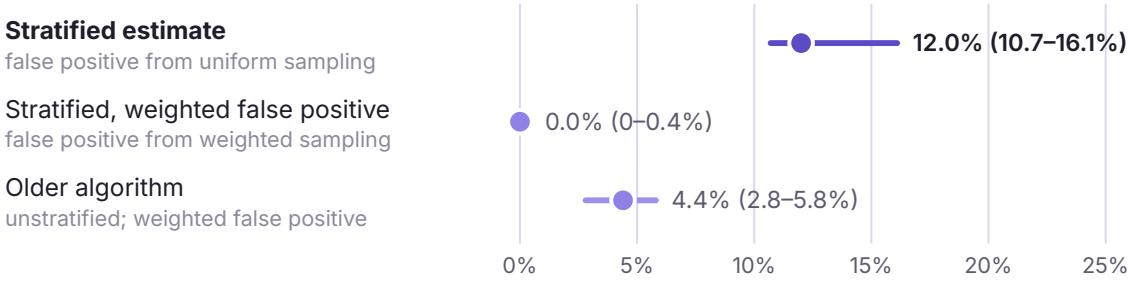
The “unknown” analysis set is about 7.9–9.4% of retained reads. Neither of two methods with different bases supports a host majority. Stratified genome alignment estimates about 12%, all of it from the 600–1,500 bp stratum with ORF, and the estimate depends on how the false positive is taken (0 with fragment-weighted non-plant genes). The expression-profile EM point estimates are 13% with ORF and 42% without ORF (calibration shows they are biased high), with many fragments of uncertain origin.

- **Scope**: the formal assembly has 96,891 genes with no protein or nucleotide hit and ≥ 10 fragments, which make up root 9.4% and leaf 7.9% of retained reads. Stratified by ORF and length, fragments without ORF are root 63% and leaf 61% of unknown fragments.
- **Composition**: median GC of unknown sequences is 41.0% (plant 44.7%, non-plant 42.7%); low-complexity fragments are only 0.4%, so these are not simple repeats.
- **Alignment to the *Cladopus* genome** (a related species in the same family [35]): stratified by ORF × length, each stratum is estimated as (unknown hit rate − false positive) ÷ (sensitivity − false positive); the host is about 12.0% of unknown fragments (bootstrap 95% interval 10.7–16.1%; 12.0%, 10.7–15.6% when the no-ORF sensitivity uses plant genes without ORF).
- **Where the 12% comes from**: all from the 600–1,500 bp stratum with ORF (hit rate 29%, sensitivity 79%, false positive 2%; about 34% of the unknown fragments pooled over the nine libraries). In the other five strata the unknown hit rate is no higher than the false positive, so they give 0, and the two non-coding controls give the same result. The false positive uses uniformly sampled non-plant genes; with fragment-weighted sampling (42% in that stratum) every stratum gives 0 (0–0.4%). The older unstratified algorithm gives 4.4% (2.8–5.8%); the difference comes mainly from stratification and the choice of false positive. Intervals reflect only sampling variation under each model.

Share aligning to the *Cladopus* genome



Host share of unknown fragments (bootstrap 95% interval)



Source Report 4.5.7, Tables 85–90

Supplement S12

“Unknown” sequences: EM estimate and one-by-one check

Method Two-class EM on the expression profiles of plant and non-plant genes, calibrated with simulated mixtures; the top 200 genes checked one by one.

EM on unknown genes, using the two-dimensional expression profiles of plant and annotated non-plant genes (log2 whole plant / mean of root and leaf × log2 root / leaf); fitted by gene and summarized with fragment weighting (host-like by gene: 3% with ORF, 21% without ORF):

Unknown fragments	With ORF	Without ORF
EM point estimate, “host-like” (by fragment)	13%	42%
Fragments with posterior ≥ 0.8	0%	8%
Fragments with posterior ≤ 0.2	81%	37%

- The rest lie in between. Calibration with simulated mixtures shows the EM point estimates are biased high (true 0.2 → estimated 0.26–0.30; true 0.5 → 0.57–0.62), so 42% is a model point estimate that may be too high, not a statistically validated upper bound.
- **How the two estimates relate**: stratified genome alignment gives about 12% (10.7–16.1%, depending on the choice of false positive; slide 25); expression-profile EM gives 13% with ORF and 42% without, of which high-posterior (≥ 0.8) host-like fragments are 0% and 8%. The two methods rest on different bases and assumptions and cannot be merged into one shared result of 12%; model posteriors are not confirmed sources, and fragments in the middle of the distribution remain uncertain.

One-by-one check of the 200 unknown genes with the most fragments (fragment-weighted shares)



Host-like (36 align to the *Cladopus* genome, 2 with a plant as nearest NR hit) **38** (15%)

Profile like host **15** (3%) Profile like associated organisms **77** (53%)

Undetermined **70** (29%)

- **One sequence checked separately**: DN26425 (1,121 bp, no ORF) alone is 1.86% of retained reads in leaf and 0.05% in root, strongly leaf-biased. It does not align to the *Cladopus* genome, and the expression profile calls it like associated organisms (host posterior 0.037); its strand, Poly(A) site and alignment to Simuliidae mitochondrial and nuclear genomes remain to be checked.

Source Report 4.5.7, Table 90

Supplement S13

Preselected secondary-wall genes in root and leaf

Method 46 preselected *Arabidopsis* genes → Proteinortho correspondence (annotation-name fallback); strict DESeq2 root vs leaf test.

> Several monolignol biosynthesis genes are higher in the non-root part; CESA4/7/8 were matched only by annotation name and are low in root and leaf overall; no clear counterparts were found for the five NAC master-switch targets.

Expression patterns

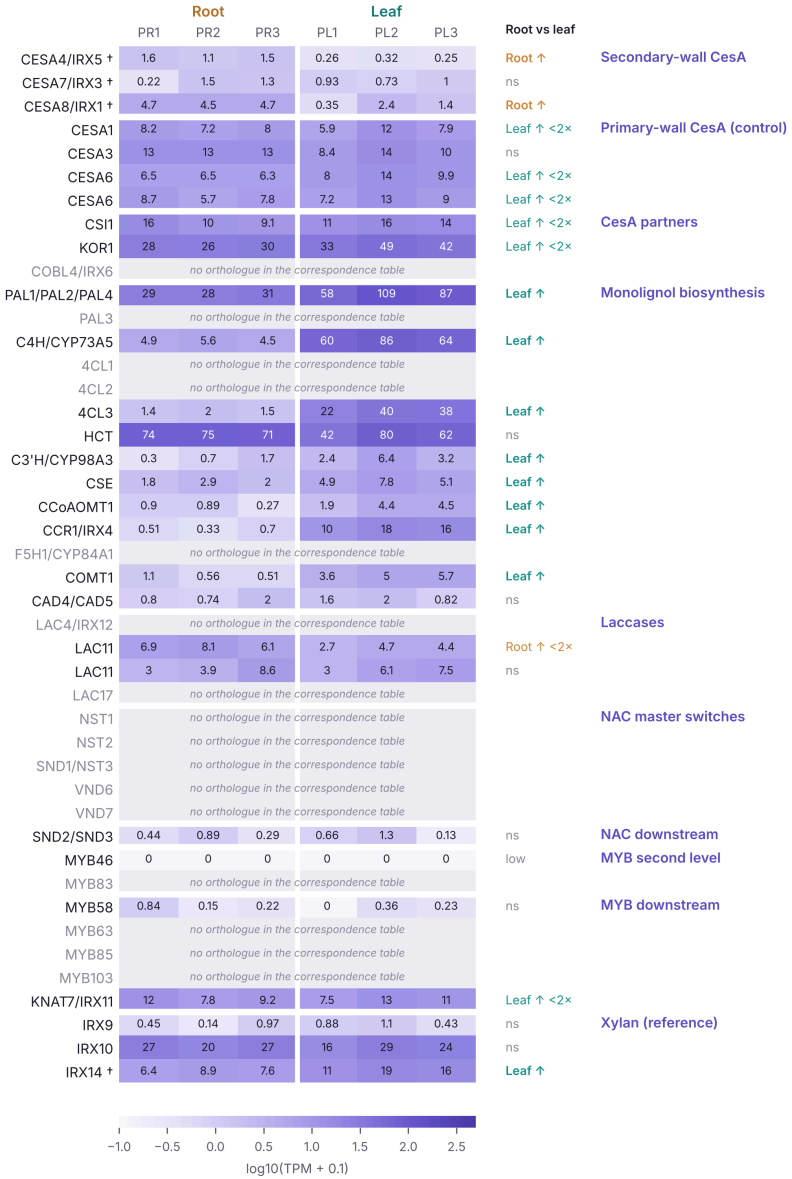
1. The upstream monolignol pathway (PAL, C4H, 4CL, CCR, CSE) is clearly higher in the non-root part than in root, with HCT as the exception;
2. The secondary-wall CESA4/7/8 (matched only by annotation name) are very low in both tissues; primary-wall CESAs, CSI1 and KOR1 are expressed normally, with root–leaf differences all under 2-fold;
3. No counterparts of the NAC master switches were found in the correspondence table; MYB46 is too low to enter testing, and MYB83 has no counterpart.

The selected typical secondary-wall module shows no coordinated high expression so far; low TPM and the lack of clear counterparts alone cannot show whether the tissue deposits secondary walls. Whether higher leaf expression of monolignol genes also serves other phenylpropanoid products is a hypothesis to test.

- NAC master switches 0 / 5 comes from searching the correspondence and strict tables by list name. All annotations also contain a 245 bp SND1 (NAC012)-like fragment (about 4 TPM, PR1 only), so low expression is one possible reason; an SND3 (NAC010)-like gene, DN69967_c1_g1, is root-high (log2FC 2.97) but not in the correspondence table. Absence, low expression and many-to-many correspondence are not yet told apart; a NAC family tree is needed (slide 29).
- Basis of selection: sections 3.2 and 5.2 of the research proposal preselected pathways related to secondary-wall thickening (CesA, monolignol biosynthesis, NAC / MYB master switches). Only the expression of preselected genes is shown, to generate hypotheses; no co-expression network is computed and no regulatory relationship is inferred.

Secondary-wall candidate genes

TPM per root and leaf library for the 46 *Arabidopsis* genes (rows sharing one *P. ceratophyllum* gene merged)
Right: strict root vs leaf test. ↑: padj < 0.05 and ≥ 2-fold; ↑ <2×: padj < 0.05 but under 2-fold; ns: padj ≥ 0.05
low: not tested (low expression). †: matched by annotation name only (weaker)



Supplement S13

Preselected secondary-wall genes: correspondence and root–leaf expression

Method 46 preselected *Arabidopsis* genes → Proteinortho correspondence (annotation-name fallback); strict DESeq2 root vs leaf test (script 45: padj < 0.05 and at least 2-fold). 30 of the 46 have counterparts in *P. ceratophyllum*, 26 from the correspondence table and 4 by annotation name only; TPM is the mean of three replicates (root / leaf).

Module	*Arabidopsis* genes	Role	Counterparts and root–leaf expression
Secondary-wall cellulose synthase	CESA4, CESA7, CESA8	three subunits of the secondary-wall cellulose synthase complex [36]	3 / 3 matched (by annotation name only, weaker), 3 genes: root-high 2, not significant 1; root and leaf TPM all below 5
Primary-wall cellulose synthase (control)	CESA1, CESA3, CESA6	primary-wall cellulose synthesis	3 / 3; all 4 genes under 2-fold (3 with padj < 0.05, slightly higher in leaf): CESA1 → DN19941_c0_g5 (8 / 9), CESA3 → DN19941_c0_g3 (13 / 11), CESA2/5/6/9 joint → DN9010_c0_g4 (6 / 11), DN9010_c0_g5 (7 / 10)
Accessory components	CSI1, KOR1, COBL4	accessory components of cellulose synthesis	2 / 3 (no counterpart for COBL4): CSI1 → DN3365_c0_g1 (12 / 14), KOR1 → DN4474_c0_g1 (28 / 42); padj < 0.05 but both under 2-fold (slightly higher in leaf)
Monolignol biosynthesis	PAL1–4, C4H, 4CL1–3, HCT, C3′H, CSE, CCoAOMT1, CCR1, F5H1, COMT1, CAD4, CAD5	phenylpropanoid pathway to monolignols [37, 49]	13 / 17, 10 genes: leaf-high 8, not significant 2. PAL1/2/4 joint → DN17_c0_g1 (29 / 84, leaf-high), C4H → DN57988_c0_g1 (5 / 70, leaf-high), 4CL3 → DN1839_c0_g1 (2 / 33, leaf-high), CSE → DN34170_c0_g1 (2 / 6, leaf-high), CCR1/2 joint → DN22401_c0_g1 (1 / 15, leaf-high); HCT → DN1840_c0_g1 (73 / 61, not significant); the other 3 leaf-high genes have root and leaf TPM below 5
Lignin polymerization	LAC4, LAC11, LAC17	laccases in lignin polymerization [38]	1 / 3 (only LAC11, under the LAC11/12/13/14/TT10 joint label): DN3493_c0_g5 (7 / 4), DN97157_c0_g1 (5 / 6); root–leaf differences both under 2-fold
NAC master switches	NST1, NST2, SND1 (NST3), VND6, VND7	master transcription factors for fiber and vessel secondary walls [39–41]	0 / 5: the name fallback in the correspondence and strict tables found none; across all annotations there is an SND1 (NAC012)-like short fragment, DN412705_c0_g1 (245 bp, low expression)
MYB second-level switches	MYB46, MYB83	direct downstream targets of the NAC master switches; redundantly regulate secondary-wall synthesis [42]	1 / 2, 1 gene, too low to enter testing
MYB downstream and others	MYB58, MYB63, MYB85, MYB103, KNAT7, SND2, SND3	downstream members of the secondary-wall regulatory network [38, 43]	SND2/3 2 / 2, 1 gene not significant (TPM < 5); an SND3 (NAC010)-like gene of the same cluster, DN69967_c1_g1, is root-high (log2FC 2.97) and not in the correspondence table. MYB downstream 2 / 5: KNAT7 → DN4839_c0_g1 (10 / 10) under 2-fold (padj 0.021), MYB58 not significant (TPM < 5)
Xylan synthesis (reference)	IRX9, IRX10, IRX14	secondary-wall xylan synthesis	3 / 3: IRX10/IRX10L → DN309442_c0_g1 (25 / 23, not significant), IRX14 → DN404525_c0_g1 (8 / 15, leaf-high; by annotation name only), IRX9 not significant (TPM < 5)

The locus IDs of the 46 *Arabidopsis* genes were checked one by one. CESA4/7/8 and IRX14 were matched by annotation name only; joint labels are kept, and one gene can appear in several rows. “No counterpart” means not found by list name in the correspondence and strict tables; across all annotations there are low-expression sequences with the same names as COBL4, 4CL1, F5H1 and LAC4. In the atlas table, MYB103 and MYB63 appear as AtMYB103 and ATMYB63; these are aliases with the same locus IDs.

Source Report 4.4.11, Table 66

Supplement S14

Open items: paired test, family trees and a few homology relationships

> The remaining questions involve targeted computation and existing records, and some are limited by the data design.

Question	Next computation or check	Effect on current conclusions
Root–leaf pairing and independence	Confirm the actual correspondence of PR1–3 and PL1–3 and whether they come from three independent plants; then update the model with the actual pairing	Recheck DE gene counts, enrichment, marker expression, secondary-wall calls and extended candidates; recheck the permutation design
Nine-species family trees	Trees for the 14 marker groups with the seven species plus *Hypericum perforatum* (Hypericaceae, sister family of Podostemaceae) and tomato, e.g. STM with KNAT1/2/6 as the KNOX family	Places *P. ceratophyllum* sequences next to orthologues of the closest sampled relative; shows whether a duplication predates or follows the split from Hypericaceae
Correspondence of a few extended candidates	Family trees, ORFs, domains, coverage and eggNOG conflicts for SAUR55 and others	Decides whether specific gene names and functional interpretations can be used
Homology resolution of key markers	FIL and PLT1/2 undecided; YAB3 with one-sided support; KNAT2/6 and HB8 with joint labels	Add evidence only for relationships that affect key interpretations
NAC master switches of the secondary wall	NAC family tree to confirm whether *P. ceratophyllum* has members of the NST / SND / VND clades	Decides whether “no counterpart found” on slide 27 can be written as “not detected”
The largest unknown sequence	Strand and Poly(A) site of DN26425, and alignment to Simuliidae genomes	Alone it is 1.9% of retained reads in leaf and needs a separate explanation

- The 742 eggNOG conflicts are checked one by one only for the few candidates that enter interpretation.

Source Report 4.1.2, Table 3; 4.3.3, Table 29; 4.3.5–4.3.6, Tables 32, 35; 4.4.8; 4.4.11; 4.5.7, Table 90; 5.1

What the current results support

> Clear expression and source clues have been obtained, interpreted at a consistent level.

1. The screened reference transcriptome and the main root–leaf expression differences are stable under read-set checks; rechecking the model after the pairing numbers and independence are confirmed is still to be done.

2. Non-plant reads are mainly animal (Simuliidae in leaf, Chironomidae in root); bacteria, fungi and algae all have source evidence. Most families and genera that meet the association criterion are root-biased, but most re-verified fragments support only broad groups, and the taxa reportable to family or genus all carry very low fragment shares (slide 23). After all non-plant genes were scanned for host isoforms, 5 mixed gene clusters contain host isoforms meeting the rule (two “bacteria”, the midge 28S cluster, a “red alga” cluster and one trace sequence), and the matching host fragments have been deducted; stability calls for bacterial, fungal and algal families are not affected.

3. Broad gene correspondence with *Arabidopsis* has been established; agreement in direction is stronger on the leaf side, and a few developmental candidates and opposite-direction candidates merit focused interpretation.

4. The “unknown” analysis set is about 7.9–9.4% of retained reads. Stratified genome alignment estimates the host at about 12% (10.7–16.1%), all from the 600–1,500 bp stratum with ORF, and 0 with a fragment-weighted false positive. The expression profile is a different basis, its results differ with and without ORF and include many uncertain fragments; the two methods are not validation with independent samples.

5. Preselected secondary-wall genes: the upstream monolignol pathway is higher in the non-root part, secondary-wall CESAs are low, and no counterparts of the NAC master switches were found; description and hypothesis only.

Source Report 5.1, Table 99

Supplement S14

Points for discussion

1. Which bacterial, fungal and algal sequences, competing hits and reference databases to prioritize, and which specific family or genus candidates still need fuller computational taxonomic evidence.

2. Which extended candidates need family-tree confirmation before entering the main results.

3. Whether the secondary-wall expression pattern can support the hypothesis in the proposal, or should stay descriptive.

Appendix A Method definitions and data quality

Item	Definition
Original library	All clean reads of each library after cleaning (before source screening); the denominator of associated-source shares
Retained library	Reads entering dc2 assembly and quantification after two rounds of source screening (including supplementary screening with Kraken 2); retained shares are root 82.09% and leaf 68.73%
Mapped fragments	Fragments assigned to transcripts by Salmon; unmapped fragments are listed separately
Gene sets	strict: the 23,969 genes with the protein-level plant label of script 44 (main analysis); plant: nonplant excluded, ambiguous and no-hit kept, 198,739; all: all 211,391
Read sets	A: screened reads (main analysis); B: all clean reads; C: screened reads plus first-round ambiguous reads
DE calls	Conventional: BH padj < 0.05 and unshrunken $\|log2FC\| \geq 1$ (strict 5,069); threshold test: H0 $\|log2FC\| \leq 1$, H1 $\|log2FC\| > 1$ (strict 2,466) [28]
Source label versions	V0: original labels of script 84; V1: nt revision of script 86; VF: after sequence-by-sequence re-verification by script 88; VFi: VF with host fragments further deducted by isoform

Assembly and quantification quality

89.9% BUSCO completeness of dc2 unigenes (embryophyta_odb10, n = 1,614; all isoforms 91.4%) [3]

≥ 95.8% mapping rate in every root and leaf library: root 95.8–95.9%, leaf 98.8–99.0% (first 2 million pairs of assembly input reads per library, Bowtie2 end-to-end)

43,013 dc2 transcripts carrying 90% of expression (127,989 in the pre-screening assembly)

- Assembly: Trinity [44, 45]; supplementary read screening: Kraken 2 [46]; quantification: Salmon [29]; differential expression: DESeq2 [28]; sequence alignment and classification: BLAST+ [47], DIAMOND [34], NCBI Taxonomy [48].
- Consistent results from different algorithms or programs on the same data are not independent validation.

Source Report 4.2.5, Table 14; 4.2.7, Tables 17–18; 4.2.10, Table 23; 4.4.1–4.4.2, Tables 36–38; 4.4.4, Table 42; 4.5.3; 4.5.8, Table 95

Appendix B Homology evidence for the 24 vegetative development targets

dc2 IDs omit the prefix TRINITY_; R_TN_ marks sequences added back by tblastn re-query. “Support”: conservative bootstrap support in the dc2 focused tree; “norm”: call in the seven-species focused tree of the pre-screening assembly; “Not detected”: not detected this time.

Marker	dc2 sequence	Support	Adopted	Basis and limits
Shoot apical meristem				
STM	DN11528_c0_g1_i3	94%	Medium	norm 82%; both medium
KNAT1	DN69988_c0_g1_i1	98%	Medium	norm 94%; lower confidence taken
KNAT2/6	DN90238_c0_g1_i1	92% / 91%	Medium, joint label	norm 100%; one sequence, counted once
WUS	DN125409_c0_g1_i1	97%	High	only ~25% of protein, around homeobox
CLV3	—	—	Not detected	CLE method recall 75% in *Arabidopsis*
Leaf				
AS1	DN70555_c0_g1_i1	100%	High	both agree
YAB3 / FIL	DN180872_c0_g1_i4	51% / 68%	YAB3 medium; FIL undecided	YAB3 one-sided support (norm 85%); one sequence, counted once
FIL/YAB3-like	DN89249_c0_g1_i3	48% / 60%	Undecided	used for neither
YAB5	DN1141_c0_g1_i1	99%	High	norm 95%
YAB2	—	nearest 0%	Not detected	beyond a duplication node
CUC1	—	nearest 8%	Not detected	DN313831 is the CUC2 candidate
CUC2	DN313831_c0_g1_i2	92%	Medium	norm 90%
CUC3 (a) / (b)	DN4024_c0_g1_i1; DN402342_c0_g1_i1	97% / 100%	High, 2 sequences	possibly two genes, listed separately

Marker	dc2 sequence	Support	Adopted	Basis and limits
Root				
SHR	DN302812_c0_g1_i1	100%	High	both agree
SCR	DN361067_c0_g1_i1	100%	High	both agree
WOX5	—	nearest 1%	Not detected	no protein-level reads
PLT1/2	candidates DN9331_c0_g1_i1, R_TN_DN10568_c0_g1_i4	0%	Undecided	other AIL in norm; listed as AIL-like
Vascular				
CLE41/44	DN121057_c0_g1_i1, DN419884_c0_g1_i1	—	TDIF-class motif candidates	CLE41 / 44 not separable
PXY	DN60_c1_g1_i1	100%	High	norm 99%
WOX4	DN127383_c0_g1_i1	100%	High	norm 99%
HB8	R_TN_DN1042_c0_g1_i80	90%	Medium	dc2 joint ATHB-15/HB8; norm 100%

- By *Arabidopsis* target: high 8, medium 7, undecided 3, not detected this time 4, TDIF-class motif candidates 2; KNAT2/6 are 2 targets and 1 sequence, CUC3 is 1 target and 2 sequences.
- “High” is limited by sampling only two species: within-family duplication, differential loss and unsampled paralogs cannot be excluded; for WUS it applies only to the fragment.
- The CLE motif method recovers 75% of the 32 *Arabidopsis* precursors overall, not a detection rate for CLV3; not detected is not evolutionary loss.

Source Report 4.3.6, Table 35

Supplement S2–S5

Appendix C Sequence-level evidence for associated sources

Method Per-sequence evidence from nt, NCBI rRNA databases and NR; common ancestor of hits within 1%.

IDs are from the first-round assembly (norm) unless marked dc2; identity and coverage are best-hit values. Reported ranks are descriptive upper limits, not species identification.

Sequence	Original label or origin	Main evidence	Reported rank or handling
DN61527_c0_g1	Bacteria	nt *Methylobacterium radiotolerans* 16S 94.2% (full length); 16S database *Rubellimicrobium roseum* 98.7% (coverage 68%); host mitochondrion 79%; i9 100% identical to host plastid, full length	Host plastid 16S rRNA; the longest non-host isoform reaches only Alphaproteobacteria
DN2273_c0_g1	Bacteria	nt *Chamaesiphon minutus* 23S 89.7% (full length); LSU database *Geitlerinema* 83.9%; host mitochondrion 87.3%; i2 99.2% to the host plastid (coverage 99%)	Host plastid 23S rRNA; the longest non-host isoform reaches order (Gomontiellales)
DN22134_c0_g2	Stramenopiles → red algae (script 86)	nt *Palmaria decipiens* 93.0% (full length); 97–99.9% of fragments on host plastid rRNA isoforms	Recorded as host; the remaining red-alga-like isoforms reach Florideophyceae
DN2745_c0_g1	*Culex* → Chironomidae (script 86)	nt *Heterotrissocladius* 28S 92.0% (coverage 0.94); fragments on the host nuclear 28S rDNA isoform: root 96.5%, leaf 73.3%	Mixed cluster of Chironomidae 28S and host nuclear rDNA
DN586224	Bacteria	family by script 88; NR best hit bacterial ammonia monooxygenase subunit AmoC, 100%; amoA, amoB, hao, nxrA, nxrB not detected	Nitrospiraceae (family)
DN1834_c0_g1	Stramenopiles	*Melosira dubia* 28S 80.4% (coverage 0.70)	Diatom-like; below the order threshold, family undetermined
DN7210_c0_g1	*Cryptosporidium* (protein level) → other eukaryotes (script 86)	nt best hit *Audouinella hermannii* (red alga) 18S 90.4% (full length); of 10 hits within 1%, 9 are red algae and 1 an environmental sequence without taxonomic information; without it the common ancestor is Florideophyceae	Red algae (Florideophyceae), reassigned on 2026-10-28
dc2 DN2239_c0_g1_i8	dc2 residue	18S best hit *Zygophlyctis* 85.6% (coverage 0.78); the common ancestor of close hits is only Eukaryota	Chytrid-like 18S, rank only Eukaryota
dc2 DN150430_c0_g2	Plant gene	newly differential with all clean reads; the source of the mis-mapped reads (125,603 pairs) has closest protein reference *Sordaria macrospora*, 78.9%	Mis-mapped fungus-like reads, not identified to species
dc2 DN99759_c0_g1	dc2 residue	0.019% of the root retained library, 439 bp, no ORF; NR best hit *Aspergillus steynii* 92.3% (2 of 5 hits fungal); ambiguous in norm, plant-labelled in dc (cmp3 explanation report, X3)	Source undetermined; best hit fungal
DN168044	Fungi → embryophytes	nt Poaceae 99.2% (full length); about 0.036% of the root original library, almost all from PR3	Non-host plant, removed from non-plant
DN50241_c0_g1	Fungi → Metazoa	nt *Philodina* 28S 99.7% (full length)	Philodinidae
DN45031_c0_g1	Metazoa	nt *Simulium ornatum* 97.7% (coverage 0.99)	Simuliidae
19 animal markers	Metazoa	black flies, midges, naidid oligochaetes, net-spinning caddisflies and others; identity ≥ 97% and coverage ≥ 0.5	Closest references and common ancestor within 1%: Supplementary Data S2

Source Report 4.4.10, Table 64; 4.5.2, Table 71; 4.5.4, Table 79; 4.5.5, Table 81; 4.5.6, Table 84; 4.5.8, Tables 92–94

Data sources

Material	Description
Report	Comprehensive Experimental Record Report of the *Podostemum ceratophyllum* Transcriptome (Podostemum_ceratophyllum_Integrated_Experimental_Record_20261026.docx; 2026-10-26, revised 2026-10-28). “Source” at the bottom of each slide gives the matching section and table numbers
Supplementary Data	Supplementary Data to these slides (2026-10-29; sections S1–S15); the tag at the top right of each slide gives the matching section. Detailed values not listed in the Report (such as the re-verified shares of each major group and the identities of reassigned sequences) are given there with their sources
Figures	fig05, 06, 07, 16, 19 and 21 redrawn on 2026-10-29 with root and leaf only, from the data tables of the display figure set of 2026-10-26; fig22 (pie chart) from the same values as fig05; fig15 unchanged (figs/ of the slide source); per-library values in Report Table 92 and Appendix B, Tables 102–104 and 106
Results records integrated in the Report	Results records 4.2, 4.2dc, 4.3 (four updates), 4.4, 4.3–4.4dc2 and 4.4–4.5 (Report appendix, Table 101)

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